

Lead Management & Enquiry Automation Platform

System Architecture, Implemented Features & Strategic Roadmap

Author: Zywo Labs Engineering Team | Date: September 2026 | Version: 1.0 (MVP Complete)

1. Executive Summary & Core System Architecture

The **Lead Management & Enquiry Automation Platform** is a production-quality, self-hosted franchise enquiry handling system built to operate 100% independently without reliance on mandatory third-party automation tools. It provides authoritative multi-tenant data isolation, strict role-based access control (RBAC), automated rule-based qualification scoring, and dynamic operational portals for both Brand Owners and Sales Teams.

Component	Technology & Pattern	Responsibility
Frontend	React, TypeScript, Vite, Tailwind CSS, Lucide	Interactive UI, responsive views, state management
Backend API	Python, Django, Django REST Framework, SimpleJWT	Authoritative business logic, qualification engine, RBAC
Database	PostgreSQL	ACID compliant relational storage & lead sequence state
Infrastructure	Docker & Docker Compose	Containerized multi-service orchestration

Mandatory Security Principle: All permissions, queryset isolation, object-level security, and qualification calculations are strictly enforced by Django REST Framework on the backend. React acts purely as a presentation layer.

2. Currently Implemented Features (MVP Status: Complete)

2.1 Multi-Tenant & RBAC Hierarchy

- **Organization & Brand Scoping:** Clear hierarchical isolation where Organizations own multiple Brands, and users/leads are scoped strictly per Brand.
- **Role-Based Access Control:** Three distinct roles configured with granular permissions:
 - **Brand Owner:** Full admin rights across leads, qualification rules, team management, analytics, and organization settings.
 - **Sales Manager:** Access to all brand leads, activity logs, and sales portal views.
 - **Sales Agent:** Scoped view restricted exclusively to leads assigned to them.
- **JWT Authentication & Flow:** Access and Refresh tokens with automatic client header injection and role-based route guards in React.

- **Organization Creation Flow:** Streamlined setup allowing new brand creation with automatic slug generation and owner initialization.

2.2 Atomic Lead Management & Lifecycle

- **Atomic Lead Numbering:** Database transactions (``select_for_update``) ensure zero collision in sequence counters (e.g. ``LEAD-000001``).
- **Comprehensive Lead Model:** Customer contact info (name, phone, email, city), preferred location, investment capacity (Decimal format), property availability, business experience, and start timeframe.
- **Sales Status Pipeline:** Complete lifecycle states: NEW, CONTACTED, FOLLOW_UP, MEETING, NEGOTIATION, CONVERTED, LOST.
- **Lead Assignment Engine:** Instant assignment to sales agents with recorded timestamp (``assigned_at``) and automatic activity log generation.

2.3 Automated Lead Qualification Engine

- **Hard Disqualification Rules:** Instant automated disqualification if key requirements are unmet (e.g. investment capacity below the minimum brand threshold, such as **■15,00,000**).
- **Weighted Point Scoring (0 to 100):** Configurable dynamic scoring engine calculating cumulative points:
 - **Investment Tier:** High (+30 points), Medium (+15 points).
 - **Location Match:** Target city/state match (+20 points).
 - **Property Ready:** Existing commercial location available (+20 points).
 - **Business Experience:** Prior retail/franchise background (+10 points).
 - **Immediate Start:** Ready to launch within 0-3 months (+20 points).
- **Qualification Outcomes:** Automated status classification into QUALIFIED (score $\geq$ threshold), REVIEW, or NOT_QUALIFIED.

2.4 Communications & Activity Audit Trail

- **Unified Timeline:** Customer communication log (Inbound/Outbound via WhatsApp, Call, Email) displayed chronologically.
- **Internal Staff Notes:** Sales managers and agents can post internal notes on lead files for team collaboration.
- **System Audit Logs (``ActivityLog``):** Immutable audit record capturing lead creation, field updates, qualification status recalculations, agent assignments, and follow-up schedules.

2.5 Dual Role-Tailored Web Portals

Feature Area	Owner Portal Capability	Sales Portal Capability
Dashboard KPI	Total leads, qualification breakdown, conversion rates, brand performance	My assigned leads, urgent follow-ups today, recent activity queue
Lead Table & Search	Filter by qualification status, lead source, assigned agent, date range	Search assigned leads, fast status toggling, call logging
Lead Detail View	Full lead parameters, score breakdown, qualification reason, activity timeline	Customer info, notes log, follow-up scheduler, quick whatsapp launch
Rule Management	Dynamic rule builder for hard disqualifiers and point weightings	Read-only qualification badge and reason display
Team & Admin	Invite/create agents, assign roles, analytics dashboard, organization settings	Personal profile management, password updates

2.6 Automated Verification & Test Suite

- **Seeded Test Environment (`seed\_data`):** Instant populated dev database with pre-configured Brand Owner (`owner@zywolabs.com`), Sales Manager (`manager@zywolabs.com`), and Sales Agents (`agent1@zywolabs.com`, `agent2@zywolabs.com`).
- **Backend Automated Test Suite:** Pytest / Django unittest suite validating authentication security, object-level queryset isolation, qualification calculation accuracy, and cross-brand privacy integrity.

3. Needed Implementations (Gap Analysis for Production)

While the core MVP application and business logic are fully functional and validated, the following key modules are required for complete enterprise deployment and automated enquiry scaling:

Category	Module / Feature	Description & Technical Requirement	Priority
Automation	WhatsApp Cloud API Integration	Inbound webhook receiver to parse incoming customer WhatsApp messages into the conversation timeline; Outbound template messaging for instant automated qualification responses.	HIGH
AI Engine	LLM Lead Parameter Extractor	OpenAI / LLM pipeline to automatically analyze raw chat transcripts and extract investment budget, city, property state, and timeline into structured Django Lead fields.	HIGH
Integrations	n8n Workflow Bridge	Webhook emission engine and n8n workflow triggers for external CRM sync, Meta Ads lead form webhooks, and automated email drip campaigns.	MEDIUM
Real-Time	Live Push Notifications	WebSocket / Server-Sent Events (SSE) notification bell alerting agents immediately when a new lead is assigned or a high-score lead registers.	MEDIUM
Reporting	CSV / PDF Export Engine	One-click CSV/Excel export for lead lists, qualification summary reports, and agent performance funnels in the Owner Portal.	NORMAL

DevOps	Production Caddy & SSL Setup	Caddy reverse proxy configuration for automated Let's Encrypt SSL/TLS certs, rate limiting, and PostgreSQL automated backup scripts.	HIGH
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4. Strategic Next Steps & Phased Execution Plan

To transition seamlessly from the current working MVP to a fully automated production system, the following 3-phase execution roadmap is recommended:

Phase 1: Platform Polish & Lead Export (Week 1)

- 1. Implement CSV / Excel Export:** Add export backend API endpoint and UI download buttons in OwnerLeads.tsx and OwnerLeadDetail.tsx.
- 2. Notification Badge System:** Add visual indicator for overdue follow-ups and unassigned leads on the Sales and Owner dashboards.
- 3. Enhanced Filters:** Add multi-select filtering by budget ranges, cities, and specific lead sources.

Phase 2: Communication & AI Automation (Weeks 2 - 3)

- 1. WhatsApp Webhook Endpoint:** Build backend/integrations/whatsapp_views.py to handle inbound WhatsApp messages, matching phone numbers to existing leads or creating new leads automatically.
- 2. AI Extract Service:** Deploy lightweight LLM parser function in backend/qualification/services.py to automatically extract structured values (investment, city, property) from raw customer text.
- 3. n8n Orchestration Workflows:** Connect n8n workflows for inbound Facebook/Instagram Ads webhook triggers directly into Django API.

Phase 3: Security Hardening & Production Deployment (Week 4)

- 1. Production Infrastructure:** Finalize infrastructure/caddy/Caddyfile with HTTPS domain routing and static asset compression.
- 2. Database Resilience:** Configure daily automated encrypted PostgreSQL backups and environment variable secret storage.
- 3. End-to-End Load & Security Audit:** Run cross-tenant penetration testing and concurrent lead creation benchmarks.

Summary & Conclusion: The Lead Management & Enquiry Automation Platform has achieved a complete, robust MVP status with all core models, security isolation, scoring logic, and UI portals operating smoothly. Proceeding with Phase 1 polish and Phase 2 integrations will deliver a market-ready automated franchise lead engine.